TAFS API Specification

Bill Inquiry & Bill Payment
for United Bank Limited (UBL)

The American Foundation School (TAFS)
TAFS IT and Software Department

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1 Overview

TAFS exposes two HTTPS APIs to UBL for collecting school fees:

- **Bill Inquiry** (`GET`): returns the student's bill and the exact amount to collect.
- **Bill Payment** (`POST`): posts a collected payment against that bill.

Both APIs take a single identifier, the `consumer_number`. Its **length** decides which amount is returned (Section 2).

2 Consumer Number

Length	Format	Amount returned
5 digits	CC number, zero-padded (`08088`)	**Normal amount**: the voucher amount, with no addition.
15 digits	10-digit TAFS 1Bill prefix + 5-digit CC (`100000000008088`)	**1Bill amount**: the voucher amount + the 1Bill charge (Section 3.2).

- A CC may be sent with or without leading zeros (`8088` = `08088`); any number of up to 5 digits is treated as a CC.
- A 15-digit number must start with the TAFS 1Bill prefix; its last 5 digits are the CC.
- Any other length, or a 15-digit number with a different prefix, returns response code `01`.
- Examples in this document use the prefix `1000000000`.

3 Amounts

3.1 Voucher amount

Payment date	Voucher amount
On or before the due date	`amount_within_due_date`
After the due date, up to the validity date	`amount_after_due_date` (includes the late payment surcharge, `late_fee`)
After the validity date	Not payable; response code `04`

Dates are evaluated in Pakistan Standard Time (UTC+05:00).

3.2 1Bill charge

For a 15-digit consumer number, the amount returned is the voucher amount plus the **1Bill charge**, which the payer bears. The 1Bill charge is the sum of the 1LINK Same Day Settlement Service Fee and the 1LINK Direct Biller / Bill Aggregator charge for the transaction slab:

Transaction slab (PKR)	Same Day Settlement	Biller Charge	1Bill Charge
Up to 5,000	4.00	12.50	**16.50**
5,000.01 to 10,000	5.00	15.50	**20.50**
10,000.01 to 100,000	14.00	39.00	**53.00**
100,000.01 to 250,000	22.00	62.50	**84.50**
250,000.01 to 1,000,000	44.00	125.00	**169.00**
1,000,000.01 to 2,500,000	88.00	250.00	**338.00**
2,500,000.01 to 5,000,000	131.00	375.00	**506.00**
Above 5,000,000	175.00	500.00	**675.00**

Slab selection. The slab is that of the **transaction amount**, i.e. the voucher amount plus the 1Bill charge. Where adding the charge moves the total into the next slab, the next slab's charge applies. The within-due-date and after-due-date amounts are each calculated separately.

Voucher amount	1Bill charge	1Bill amount	Note
90,000.00	53.00	90,053.00	Slab 10,000.01 to 100,000
91,000.00	53.00	91,053.00	Same voucher, after the due date
99,980.00	84.50	100,064.50	99,980 + 53 exceeds 100,000, so the next slab applies
4,990.00	20.50	5,010.50	4,990 + 16.50 exceeds 5,000, so the next slab applies

3.3 Collection rules

- The inquiry returns `amount_payable`: the exact amount to collect today for the consumer number sent.
- The payment must carry exactly that amount. Partial payments and overpayments are rejected (code `06`).
- On a 1Bill payment TAFS credits only the voucher amount to the student; the 1Bill charge is recorded against the payment and returned as `channel_fee`.

4 Voucher Resolution

Every inquiry and payment resolves to the student's **most recent voucher**, excluding vouchers cancelled because a newer voucher replaced them, and vouchers not yet released to the parent.

- If the student has more than one unpaid voucher, the most recent one is returned and collected first. The older one is returned only after the most recent one is paid.
- If the student has no payable voucher, the API returns the state of the most recent one (code `03`, `04` or `05`).
- Every voucher includes all arrears owed by the student.

5 Connectivity & Authentication

5.1 Base URL

```
https://tafs-backend-production.up.railway.app/api/v1/ubl
```

A separate UAT base URL and test students are provided by TAFS for testing.

5.2 Headers

Header	Type	Description
`X-UBL-User`	string	Service user ID issued to UBL
`X-UBL-Password`	string	Password for that service user
`Content-Type`	string	`application/json` (payment requests)

- Credentials are issued by TAFS separately and are never sent in the URL.
- All traffic is HTTPS (TLS 1.2 or higher). Access can be restricted to UBL's outbound IP addresses.
- Missing or wrong credentials return response code `10`.

5.3 HTTP status

- A well-formed, authenticated request always returns **HTTP 200**. The outcome is in `response_code`.
- A malformed request, such as a missing field or a non-numeric consumer number, returns **HTTP 400** with a description of the invalid field.

6 API 1 – Bill Inquiry

```
GET /api/v1/ubl/bill-inquiry?consumer_number={consumer_number}
```

6.1 Query parameters

Parameter	Type	Required	Description
`consumer_number`	digits	Yes	5-digit CC (normal amount) or 15-digit 1Bill number (1Bill amount).

6.2 Response fields

Field	Type	Description
`response_code`	string	`"00"` on success
`response_desc`	string	`"Success"`
`consumer_number`	string	As sent
`onebill_consumer_number`	string	The student's 15-digit 1Bill number
`cc`	string	Student CC, 5 digits zero-padded
`student_name`	string	Student's full name
`gr_number`	string	School registration number
`campus`	string	Campus name
`class`	string	Class
`voucher_number`	string	Voucher being collected
`billing_month`	string	Month(s) billed, e.g. `"OCT 26"`
`voucher_status`	string	`UNPAID`, `OVERDUE` or `PARTIALLY_PAID`
`issue_date`	string	`YYYYMMDD`
`due_date`	string	`YYYYMMDD`
`validity_date`	string	`YYYYMMDD`; not payable after this date
`currency`	string	`"PKR"`
`amount_within_due_date`	number	Payable on or before the due date (incl. 1Bill charge for a 1Bill number)
`amount_after_due_date`	number	Payable after the due date (incl. late fee, and 1Bill charge for a 1Bill number)
`late_fee`	number	Late payment surcharge after the due date
`channel_fee`	number	1Bill charge included in `amount_payable`; `0` for a CC
`amount_payable`	number	**Exact amount to collect today**

Amounts are in PKR with up to two decimal places.

6.3 Sample – CC (normal amount), after the due date

```
GET /api/v1/ubl/bill-inquiry?consumer_number=08088
X-UBL-User: <issued by TAFS>
X-UBL-Password: <issued by TAFS>
```


```
{
  "response_code": "00",
  "response_desc": "Success",
  "consumer_number": "08088",
  "onebill_consumer_number": "100000000008088",
  "cc": "08088",
  "student_name": "HAFSA DAHRI",
  "gr_number": "7001",
  "campus": "Gulistan-e-Jauhar Campus",
  "class": "AS",
  "voucher_number": "12461",
  "billing_month": "OCT 26",
  "voucher_status": "OVERDUE",
  "issue_date": "20261001",
  "due_date": "20261010",
  "validity_date": "20261020",
  "currency": "PKR",
  "amount_within_due_date": 90000,
  "amount_after_due_date": 91000,
```

```
  "late_fee": 1000,
  "channel_fee": 0,
  "amount_payable": 91000
}
```

6.4 Sample – 1Bill number (1Bill amount), before the due date

```
GET /api/v1/ubl/bill-inquiry?consumer_number=100000000008088
```

```
{
  "response_code": "00",
  "response_desc": "Success",
  "consumer_number": "100000000008088",
  "onebill_consumer_number": "100000000008088",
  "cc": "08088",
  "voucher_number": "12461",
  "voucher_status": "UNPAID",
  "amount_within_due_date": 90053,
  "amount_after_due_date": 91053,
  "late_fee": 1000,
  "channel_fee": 53,
  "amount_payable": 90053,
  ...
}
```

6.5 Sample – failure

A failed inquiry returns `response_code`, `response_desc` and, where a voucher was found, `voucher_number`:

```
{ "response_code": "03", "response_desc": "Voucher is already paid", "voucher_number
    ": "12461" }
```

7 API 2 – Bill Payment

```
POST /api/v1/ubl/bill-payment
Content-Type: application/json
```

7.1 Request body

Field	Type	Required	Description
`consumer_number`	digits	Yes	Same value used in the inquiry
`amount`	number	Yes	Amount collected, PKR. Must equal `amount_payable`
`channel`	enum	Yes	`OTC`, `APP`, `INTERNET_BANKING` or `ONEBILL`. Must be `ONEBILL` for a 15-digit number
`payment_method`	enum	Yes	See Section 7.2
`transaction_id`	string	Yes	UBL's unique transaction reference (max 100 characters)

Field	Type	Required	Description
`transaction_datetime`	ISO 8601	Yes	With offset, e.g. `2026-10-05T11:00:00+05:00`
`voucher_number`	string	Advised	From the inquiry; rejects the payment (code `07`) if a newer voucher was issued since
`branch_code`	string	OTC	UBL branch code
`payer_bank_code`	string	1Bill	Bank or wallet the payer paid from
`payer_bank_name`	string	1Bill	Name of that bank or wallet
`instrument_number`	string	Cheque / PO	Required when `payment_method` is `CHEQUE` or `PAY_ORDER`

7.2 Payment method

`payment_method`	Meaning
`CASH`	Cash at a branch counter
`CHEQUE`	Cheque (send `instrument_number`)
`PAY_ORDER`	Pay order / demand draft (send `instrument_number`)
`ACCOUNT_TRANSFER`	Debit from the payer's bank account
`DEBIT_CARD`	Debit card
`CREDIT_CARD`	Credit card
`WALLET`	Mobile wallet
`OTHER`	Any other method

7.3 Response fields

Field	Type	Description
`response_code`	string	`"00"` on success
`response_desc`	string	`"Success"`
`transaction_id`	string	As sent
`tafs_receipt_id`	string	TAFS receipt number
`voucher_number`	string	Voucher the payment was applied to
`amount_booked`	number	Amount credited to the student (excludes the 1Bill charge)
`channel_fee`	number	1Bill charge in the amount collected; `0` for a CC

7.4 Sample – 1Bill payment

```
POST /api/v1/ubl/bill-payment
{
  "consumer_number": "100000000008088",
  "amount": 90053,
  "channel": "ONEBILL",
  "payment_method": "ACCOUNT_TRANSFER",
  "transaction_id": "UBL20261005110000123",
  "transaction_datetime": "2026-10-05T11:00:00+05:00",
  "voucher_number": "12461",
  "payer_bank_code": "HBL",
  "payer_bank_name": "Habib Bank Limited"
```

```
}
```

```
{
  "response_code": "00",
  "response_desc": "Success",
  "transaction_id": "UBL20261005110000123",
  "tafs_receipt_id": "48215",
  "voucher_number": "12461",
  "amount_booked": 90000,
  "channel_fee": 53
}
```

7.5 Sample – OTC cheque payment after the due date

```
{
  "consumer_number": "08088",
  "amount": 91000,
  "channel": "OTC",
  "payment_method": "CHEQUE",
  "instrument_number": "00412387",
  "branch_code": "0457",
  "transaction_id": "UBLOTC0457202610121015",
  "transaction_datetime": "2026-10-12T10:15:00+05:00",
  "voucher_number": "12461"
}
```

7.6 Sample – failures

```
{ "response_code": "06", "response_desc": "Amount does not match the payable amount
    ", "amount_payable": 90053 }
{ "response_code": "08", "response_desc": "Duplicate transaction; already posted",
  "transaction_id": "UBL20261005110000123", "tafs_receipt_id": "48215" }
```

7.7 Duplicates, retries and timeouts

- Posting is idempotent on `transaction_id`. On a timeout, resend the **same** request with the **same** `transaction_id`.
- Code `08`: the payment is already recorded; treat as success. It is never booked twice.
- Code `99`: nothing was booked; retry.

8 Response Codes

Code	Meaning	Used by
00	Success	Inquiry, Payment
01	Consumer number not found or invalid length	Inquiry, Payment
02	No voucher has been issued for this student	Inquiry, Payment
03	Voucher is already paid	Inquiry, Payment
04	Voucher validity has expired	Inquiry, Payment
05	Voucher is not payable (written off)	Inquiry, Payment
06	Amount does not match the payable amount	Payment
07	A newer voucher has been issued; inquire again	Payment
08	Duplicate transaction; already posted (treat as success)	Payment
09	Invalid request (e.g. cheque without `instrument_number`, or a 15-digit number with a channel other than `ONEBILL`)	Payment
10	Invalid credentials	Inquiry, Payment
99	General error; nothing booked, retry later	Inquiry, Payment